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The $r = 2$ Law of Mantissa-Bit Bias in Trained Neural Weights

An empirical regularity for floating-point bit positions of SGD-trained networks

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Abstract

We report a quantitative regularity in the bit-level statistics of trained neural network weights. For a fixed floating-point format with M mantissa bits, the bias b_i of mantissa bit i above the training noise floor obeys $b_i \approx B_0 \cdot 2^{i-M+1}$, where $B_0 = \log_2(3) - 3/2 \approx 0.08496$ is a format-independent constant. The ratio of consecutive bit biases is therefore approximately two. We verify this regularity on BF16 weights of Mistral-7B over bit positions $m3$ through $m6$, where the measured ratio is $r = 2.014$ with coefficient of determination $R^2 = 0.9997$, and on FP32 weights of GPT-2 XL over bit positions $m11$ through $m22$, where the measured ratio is $r = 1.995$ with $R^2 = 0.9998$. Bits below the noise floor are statistically indistinguishable from a fair coin and are excluded from

the fit. We derive the law analytically from Benford's law applied to the IEEE 754 mantissa interval $[1, 2)$, recover the constant B_0 from the most-significant mantissa bit split at $3/2$, and report the exact theoretical sequence b_i , which converges to the geometric law $r = 2$ from below. The claim is scoped to the tested networks, formats, and bit positions; it is not a universal property of all numerical data, and we do not claim it holds below the noise floor or for untrained or freshly initialised weights.

Keywords: trained neural weights; floating-point mantissa; bit-level bias; Benford's law; Shannon entropy; lossless compression

This is an approved public release of a Forlais Group academic paper.

Methodology Disclosure Boundary

This paper reports the public-safe statement of the $r = 2$ law observed in the FCN research programme. It reports the analytical derivation, the measured ratios, the regression statistics, and the limitations of the claim. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

Lossless compression of trained neural network weights depends on what is and is not predictable in the bit-level statistics of those weights. Within the Forlais Compression Notation (FCN) research programme, we study these statistics directly. The FCN evidence register records a particular regularity that recurs across architectures and floating-point formats: the bias of each mantissa bit, taken with respect to a fair coin, is consistent with a geometric progression in the bit index, with common ratio approximately two.

This paper states and verifies that regularity. We report it as the $r = 2$ law. The contribution is empirical and theoretical. We give the analytical form, the measured ratios on two public model artefacts, and the derivation that links the law to Benford's law on the IEEE 754 mantissa interval.

We treat the result as a fundamental property of trained-weight distributions rather than as a property of any single codec or implementation. The headline measurements, sourced from the FCN evidence register, are a fit ratio of $r = 2.014$ with $R^2 = 0.9997$ on BF16 weights of Mistral-7B, a fit ratio of $r = 1.995$ with $R^2 = 0.9998$ on FP32 weights of GPT-2 XL, and a closed-form constant $B_0 = \log_2(3) - 3/2$ that fixes the absolute scale of the bias sequence.

The remainder of the paper states the public-safe scope, summarises the relevant background, presents the verification protocol, gives the theoretical derivation, presents the empirical results, surveys the connection to the public literature, and records the limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- the bit-level statistics of mantissa bits in trained neural network weights stored in IEEE 754 binary formats;
- the asymptotic regime where the bit index is above the training noise floor;
- public model artefacts at fully-trained checkpoints.

The result does not concern:

- untrained, freshly initialised, or partially trained networks, whose bit-level statistics are determined by the initialisation distribution and not by the multiplicative dynamics of training;
- bit positions below the training noise floor, where the measured bias is dominated by stochastic optimisation noise and is statistically indistinguishable from a fair coin;

- sign-bit or exponent-bit statistics, which have a different origin and are reported in separate FCN entries;
- lossy regimes such as quantisation-aware training, distillation, or pruning, which alter the distribution under study;
- cross-model delta regimes, where mantissa statistics are modified by the conservatism of fine-tuning and are reported separately;
- architectures that do not store weights in a standard IEEE 754 binary format.

The numerical results in this paper should not be applied to data outside the tested setting without a separate justification.

3 Background and related work

Shannon’s source coding theorem gives a lower bound on the expected code length of any lossless code for a discrete memoryless source in terms of the entropy of the source distribution [17, 3]. For neural network weights stored in a fixed floating-point format, that bound is determined by the joint distribution of sign, exponent, and mantissa bits. The structure of this joint distribution therefore controls the achievable lossless compression ratio.

The IEEE 754 standard [9] encodes a value v in a binary floating-point format as

$$v = (-1)^s \cdot 2^{e-\text{bias}} \cdot (1 + m/2^M), \quad (1)$$

where s is the sign bit, e is the biased exponent, m is the integer mantissa, and M is the number of mantissa bits. The mantissa m ranges over $\{0, 1, \dots, 2^M - 1\}$, and the significand $1 + m/2^M$ ranges over $[1, 2)$. BF16 [11, 13] sets $M = 7$; FP32 [9] sets $M = 23$.

The distribution of leading significant digits in numbers drawn from many natural processes is governed by Benford’s law [14, 2, 15, 7]. In base two, Benford’s law states that the significand $s \in [1, 2)$ has density

$$f(s) = \frac{1}{s \ln 2}. \quad (2)$$

For neural network weights trained by stochastic gradient descent, momentum, or Adam [12, 16], the update is multiplicative in the mantissa domain. Multiplicative processes have long been known to produce significands that follow Benford’s law; this is the content of the scale-invariance and product-of-random-variables theorems associated with Pinkham, Hill, and others [15, 7].

The literature on lossless compression of trained weights is recent. DFloat11 [6] reports compression of trained large language model weights by exploiting the empirical concentration of the exponent field. Earlier neural network compression work [5, 8, 4] primarily addresses lossy regimes such as pruning, quantisation, and distillation. The present paper concerns only the lossless setting and only the mantissa field; the exponent field is studied separately within the FCN research programme.

4 Verification protocol

We measure the bias of each mantissa bit i as

$$b_i = P(\text{bit } i = 1) - \frac{1}{2}, \quad (3)$$

estimated by the empirical bit frequency over the full set of stored weight values of a trained model in its native floating-point format. We restrict to weights stored in BF16 and FP32, examine each mantissa position independently, and report a single bias value per position per model.

For each format we identify the training noise floor, defined as the lowest bit index above which $|b_i|$ is detectably distinguishable from zero given the sample size. Below the noise floor, $|b_i|$ is consistent with fair-coin statistics and is excluded from the geometric fit. We fit

$$\log_2 |b_i| = \log_2 |B_0| + (i - M + 1) \quad (4)$$

by ordinary least squares over the bit positions above the noise floor and report the fitted slope ratio $r = 2^{\text{slope}}$ together with the coefficient of determination R^2 .

A measurement is counted as supporting the law only if the regression yields $R^2 \geq 0.99$ and the fitted ratio lies within ± 0.05 of two. We report two independent measurements drawn from public-model artefacts in two different formats.

5 Theoretical analysis

We derive the law from Benford's law applied to the IEEE 754 mantissa interval $[1, 2)$.

5.1 Probability mass over the mantissa

Under the Benford density $f(s) = 1/(s \ln 2)$, the probability that the mantissa integer takes value $m \in \{0, 1, \dots, 2^M - 1\}$ is

$$P(m) = \int_{1+m/2^M}^{1+(m+1)/2^M} \frac{ds}{s \ln 2} = \log_2 \left(\frac{2^M + m + 1}{2^M + m} \right). \quad (5)$$

5.2 Most-significant mantissa bit bias

The most-significant mantissa bit, bit $M - 1$, separates the significand into the lower half $[1, 3/2)$ and the upper half $[3/2, 2)$. The probability that the most-significant mantissa bit is zero is

$$P(\text{bit } M - 1 = 0) = \int_1^{3/2} \frac{ds}{s \ln 2} = \log_2 \frac{3}{2} = \log_2 3 - 1. \quad (6)$$

The bias of the most-significant mantissa bit, defined as the deviation of $P(\text{bit} = 1)$ from $1/2$, is therefore

$$b_{M-1} = (2 - \log_2 3) - \frac{1}{2} = \frac{3}{2} - \log_2 3 = -B_0, \quad (7)$$

with $B_0 = \log_2 3 - 3/2 \approx 0.084962500721156$. The sign of b_{M-1} indicates a slight excess of zeros at the most significant mantissa bit under Benford. The magnitude $|B_0|$ is the fundamental scale of the bit-level bias.

5.3 General product formula

For lower mantissa bit position i , the bit toggles at every step of size 2^{i+1-M} across the significand interval. Aggregating Benford mass over the relevant sub-intervals gives

$$b_i = \frac{1}{2 \ln 2} \ln \left(\prod_{j=N}^{2N-1} \frac{(2j+1)^2}{4j(j+1)} \right), \quad (8)$$

where $N = 2^{M-1-i}$. This is the exact closed-form sequence of mantissa-bit biases under Benford.

5.4 Asymptotic recovery of $r = 2$

For large N , that is for low bit positions i well below the most significant mantissa bit, each factor in the product satisfies

$$\frac{(2j+1)^2}{4j(j+1)} = 1 + \frac{1}{4j(j+1)}. \quad (9)$$

Telescoping the sum of small corrections,

$$\sum_{j=N}^{2N-1} \frac{1}{4j(j+1)} = \frac{1}{4} \left(\frac{1}{N} - \frac{1}{2N} \right) = \frac{1}{8N}, \quad (10)$$

gives

$$b_i \approx \frac{1}{16N \ln 2} = \frac{2^{i-M+1}}{16 \ln 2}. \quad (11)$$

Identifying the prefactor with $|B_0|$ yields

$$b_i \approx B_0 \cdot 2^{i-M+1}, \quad b_{i+1}/b_i \rightarrow 2, \quad (12)$$

with convergence from below as i decreases below the most significant mantissa bit. The convergence is rapid, as quantified in the next subsection.

5.5 Exact consecutive ratios under Benford

Table 1 reports the exact Benford ratios b_{i+1}/b_i for the BF16 mantissa under $M = 7$.

Pair	Exact ratio	Deviation from 2
B_6/B_5	1.9178	−0.082
B_5/B_4	1.9771	−0.023
B_4/B_3	1.9935	−0.007
B_3/B_2	1.9984	−0.002
B_2/B_1	1.9996	−0.0004

Table 1: Exact ratios between adjacent Benford mantissa-bit biases for BF16. The ratio approaches two from below as the bit index decreases.

5.6 Why the asymptotic ratio is exactly two

The factor-of-two convergence is a consequence of the dyadic structure of binary mantissa bits. Halving the bit index doubles the number of toggling sub-intervals of the significand, and each pair of adjacent sub-intervals contributes Benford mass proportional to a leading term that scales as $1/N$. The bookkeeping of these contributions gives a sequence of $\log_2 |b_i|$ values that is arithmetic in i with unit increment, so $|b_i|$ is geometric in i with ratio two. The exponential of the residual telescoping correction is what produces the small downward deviations recorded in Table 1.

5.7 Independence of the law from the weight standard deviation

The law is independent of the standard deviation σ of the weight distribution. Under a change of scale $W \rightarrow \alpha W$ with $\alpha > 0$, every weight value is multiplied by α , and the IEEE 754 representation absorbs this multiplication into the exponent field while leaving the significand distribution unchanged.



The mantissa-bit biases b_i therefore do not depend on α . This is the bit-level analogue of the scale-invariance theorem for Benford's law due to Pinkham [15]: any distribution whose significand is log-uniform on $[1, 2)$ admits the same set of mantissa-bit biases regardless of its overall scale.

The empirical FCN evidence corroborates this analytical prediction. Scanning σ across three orders of magnitude on Gaussian sources gives a measured most-significant mantissa bit bias that oscillates around $|B_0|$ within the sampling noise, with the median value within one per cent of the Benford prediction. The independence holds for trained-weight distributions as well, where the per-tensor σ varies widely across roles and layers without affecting the universal mantissa-bit bias sequence.

5.8 Independence from the floating-point format

The closed-form derivation depends only on the binary mantissa structure and not on the exponent width, the bias, or any other format-specific parameter. The same sequence $b_i = B_0 \cdot 2^{i-M+1}$ governs BF16 ($M = 7$), FP32 ($M = 23$), FP16 ($M = 10$), and any other IEEE 754-style binary format that stores values as $(-1)^s \cdot 2^{e-\text{bias}} \cdot (1 + m/2^M)$. The format-independence is a direct consequence of the fact that the Benford density is defined on the significand interval $[1, 2)$, which is the same interval for every binary format.

The empirical evidence corroborates this prediction. Both the BF16 measurement on Mistral-7B and the FP32 measurement on GPT-2 XL recover the same most-significant mantissa bit bias of approximately $|B_0|$, even though their mantissa widths differ by a factor of more than three. The geometric law holds in both formats with measured ratios consistent with two.

5.9 Distinguishing the law from a Gaussian baseline

A common alternative hypothesis is that the bit-level biases are simply those of a Gaussian distribution with the empirical σ and mean zero. We address this hypothesis directly. The bit-level biases of a pure Gaussian under the IEEE 754 representation are, to leading order, the same as those of the Benford distribution because a sufficiently mixed Gaussian (over scale, or across tensor roles) has a log-uniform significand. The Gaussian baseline therefore predicts the same $r = 2$ law as Benford, not a different one. The two hypotheses are distinguishable only via the residual deviations from a pure geometric law, which are present in both but with slightly different magnitudes. The dominant prediction (the $r = 2$ law itself) is what we verify.

A genuine alternative, however, is a uniform weight distribution. The uniform distribution on $[-1, 1]$ has a significand distribution that is not log-uniform on $[1, 2)$ (it is uniform on the linear scale, not the logarithmic one), and its mantissa bits are statistically unbiased. The empirical bit-level bias for a uniform weight source is zero, which is the most direct way to falsify the $r = 2$ law. We verify that trained networks satisfy the law and that uniform random data does not.

6 Empirical results

We measured $|b_i|$ directly on two public-model artefacts in two formats. Above the noise floor, the measured bit-level bias is consistent with the analytical prediction.

The training noise floor sits at approximately bit $m2$ for BF16 and at approximately bit $m10$ for FP32. Below these positions the measured bias is consistent with a fair coin to within sampling noise, so the geometric law does not apply. Above the noise floor, BF16 admits approximately five bit positions that carry detectable bias and FP32 admits approximately twelve.

Format	Model	Fit range	Measured r	R^2
BF16	Mistral-7B	$m3 - m6$	2.014	0.9997
FP32	GPT-2 XL	$m11 - m22$	1.995	0.9998
Theory	Benford limit	all bits above floor	2.000	exact

Table 2: Measured and theoretical ratios for the $r = 2$ law. The fit range excludes bit positions at or below the training noise floor.

6.1 Diagnosing earlier discordant measurements

Earlier internal measurements within the FCN evidence register produced fitted ratios of $r \approx 2.598$ for BF16 and $r \approx 1.628$ for FP32. Both values arose from including bit positions below the noise floor, where the bias is dominated by stochastic optimisation noise rather than by the multiplicative training dynamics. After restricting the regression to bit positions above the noise floor, the measured ratios collapse onto the theoretical value of two, as recorded in Table 2. The earlier numbers should be interpreted as artefacts of fit range and not as competing measurements of the law.

7 Cross-architecture evidence

The constant $B_0 = \log_2 3 - 3/2$ predicts an absolute scale for the most-significant mantissa bit bias. The FCN evidence register records measurements of $|B_0|$ on the most-significant mantissa bit across a range of public-model artefacts. The measured values agree with the theoretical value within a few per cent, with the residual gap attributable to finite training, finite sampling, and small departures from a pure Benford significand distribution. Each model studied is a mature, fully-trained checkpoint released by its developers [10, 18, 20, 1, 19].

Source	Most-significant mantissa bit $ b_{M-1} $
Theoretical Benford limit	0.084962500721
BF16 Mistral-7B (203 tensors)	0.083641
FP32 GPT-2 XL (192 tensors)	0.085264
Laplace zero-mean (analytical)	0.084965
Uniform on $[-1, 1]$ (analytical)	0.000000

Table 3: Measured magnitude of the most-significant mantissa bit bias for two public-model artefacts, compared to two analytical reference distributions. The Benford-compliant Laplace distribution matches the theoretical constant; the non-Benford uniform distribution does not.

We note that the uniform distribution is not Benford-compliant in $[1, 2)$ in the binary-mantissa sense and therefore yields a zero most-significant mantissa bit bias. This is consistent with the derivation: the $r = 2$ law is not a universal property of all numerical data but is specifically a property of distributions that approximate Benford on the binary mantissa interval.

8 Falsification tests and alternative hypotheses

8.1 Tested alternatives

We tested four alternative hypotheses against the $r = 2$ law on the public-model artefacts and the analytical distributions in our evidence register.

- **Hypothesis A: fixed-ratio law with r different from two.** Under hypothesis A, the bit-bias sequence is geometric with a common ratio $r \neq 2$. The hypothesis is falsified by the empirical regression, which finds r within ± 0.05 of two on both formats with $R^2 > 0.99$.
- **Hypothesis B: no detectable bias above the noise floor.** Under hypothesis B, the trained-weight distribution does not exhibit a structured bias sequence and all mantissa bits behave as fair coins. The hypothesis is falsified by the directly measured non-zero biases at the upper mantissa bits ($|b_5|, |b_6|$ of order 10^{-2} on BF16 and $|b_{20}|, |b_{21}|, |b_{22}|$ of order 10^{-2} on FP32).
- **Hypothesis C: bias sequence proportional to a power law other than 2^i .** Under hypothesis C, the bias sequence follows $|b_i| \propto i^\alpha$ or $|b_i| \propto e^{\beta i}$ with $e^\beta \neq 2$. The hypothesis is ruled out by the asymptotic agreement between the geometric law and the Benford closed form across both formats.
- **Hypothesis D: Benford-compliance but with an architecture-dependent prefactor.** Under hypothesis D, the asymptotic ratio is two but the prefactor $|B_0|$ varies with architecture. The hypothesis is consistent with the few-per-cent softening observed across public models, but the softening is small enough to be attributable to finite training rather than to architecture-specific variation. We do not in this paper distinguish hypothesis D from the universal Benford prediction; the data are consistent with both.

8.2 Robustness of the regression

The empirical regressions reported in Table 2 are robust to the choice of fit range above the noise floor. Restricting the BF16 regression to bit positions $m4$ through $m6$ instead of $m3$ through $m6$ gives a fitted ratio of $r = 2.02$ with $R^2 = 0.9998$; restricting it further to $m5$ through $m6$ gives a fitted ratio of $r = 1.95$. The variation is within the expected range given the small number of bits above the noise floor. The FP32 regression has more bits above the noise floor and is therefore more robust, with the ratio remaining within ± 0.01 of two across any contiguous fit range of four or more bit positions.

8.3 Sensitivity to the noise-floor threshold

The noise floor is defined operationally as the lowest bit index at which $|b_i|$ exceeds the detection threshold. The threshold itself depends on the sample size and the regression diagnostic (we use $R^2 \geq 0.99$). Shifting the noise floor by one bit position changes the fit range but does not change the fitted ratio by more than the variations recorded above. The law is therefore stable to reasonable choices of the noise-floor threshold.

9 Connection to public literature

The $r = 2$ law sits at the intersection of three public threads.

First, Benford's law has a century-long history in mathematical statistics [14, 2, 15, 7]. The translation from base ten to base two is direct: the significand density $f(s) = 1/(s \ln 2)$ over $[1, 2)$ is the binary form of the law and is what governs the mantissa of an IEEE 754 binary float. The constant $\log_2 3 - 3/2$ has appeared implicitly in the literature wherever the binary Benford density is integrated over a half-interval, but to our knowledge it has not previously been identified as a universal bit-bias constant for trained neural network weights.

Second, recent neural compression work [6] reports that the exponent field of trained large language model weights is concentrated and therefore compressible. The complementary observation about the



mantissa is that its bit-level statistics are tightly constrained by Benford, leaving only the small bias sequence captured by the $r = 2$ law. Together these two facts explain why the achievable lossless ratio for a single-model BF16 weight tensor is essentially set by exponent concentration: the mantissa contains very little additional structure to exploit.

Third, the optimisation literature on stochastic gradient descent, momentum, and Adam [16, 12] establishes that the per-step update is multiplicative in the mantissa domain. This is the mechanism that drives a fully-trained network's mantissa distribution towards Benford and is therefore the proximate cause of the bias sequence we measure. The convergence is asymptotic; freshly initialised networks do not exhibit the law.

10 The role of multiplicative training dynamics

10.1 Why training drives the weights to Benford

The mechanism by which a trained network's weight distribution approaches the binary Benford density is the multiplicative structure of standard optimisers. SGD [16] updates a parameter w by $w \leftarrow w - \eta g$, where η is the learning rate and g is the gradient. Momentum-based optimisers accumulate gradients via running averages, and adaptive optimisers such as Adam [12] normalise updates by running averages of squared gradients. Each per-step update is small relative to the parameter, so the cumulative effect over many steps is multiplicative: $w \approx w_0 \cdot \prod_t (1 + \delta_t)$ for small per-step increments δ_t .

The product of many independent random factors of the form $(1 + \delta_t)$ produces a log-normal distribution in the limit of large t . Log-normal distributions over a sufficiently large range are well-approximated by Benford on the binary mantissa [15, 7]. Trained networks at convergence therefore exhibit weight distributions whose significands approach Benford, and the empirical bit-level biases approach the closed-form Benford sequence.

10.2 Why freshly initialised networks do not exhibit the law

A freshly initialised network has weights drawn from a single-step distribution (typically Gaussian or uniform with width derived from a fan-in/fan-out factor). Such a distribution is not the product of many random factors and does not in general have a log-uniform significand on $[1, 2)$. The most-significant mantissa bit bias of a freshly initialised network is therefore close to zero, not $|B_0|$, and the $r = 2$ law does not hold.

The empirical FCN evidence corroborates this prediction. Untrained Xavier-initialised tensors and uniform-initialised tensors both show negligible most-significant mantissa bit bias at the precision of the measurement. The bias appears only after sufficient training steps to drive the weights through enough multiplicative compounding to reach the asymptotic Benford regime.

11 Implications

The law has three implications for trained-weight compression.

First, the mantissa field of a single-model trained weight tensor carries close to maximal Shannon entropy. The total information saved by exploiting the $r = 2$ law on BF16 mantissas is of order 0.03 bits per value out of seven mantissa bits, which is below one per cent of the mantissa entropy and below half a per cent of the total sixteen-bit entropy. The mantissa is therefore a poor target for additional lossless compression effort in the single-model setting.



Second, the absolute scale $|B_0|$ is set by Benford and is independent of the architecture, the standard deviation of the weight distribution, or the floating-point format. This explains why the empirical compression ceiling for single-model BF16 weights is stable across architectures: the mantissa contribution is universal.

Third, departures from the $r = 2$ law are diagnostic. A trained model whose mantissa bits deviate significantly from the law has either been trained under non-standard dynamics, has been quantised or otherwise post-processed, or has its bit-level statistics dominated by sub-noise-floor positions. The law therefore serves as a sanity check on the bit-level fidelity of any weight artefact that claims to be trained by standard multiplicative optimisation.

12 Operational significance for FCN evidence

12.1 The law as a sanity check

Within the FCN evidence register, the $r = 2$ law functions as a sanity check on the bit-level fidelity of trained-weight measurements. A measurement that fails to recover the law at the upper mantissa bits suggests either that the network has not been trained to convergence, that the artefact has been post-processed (for example by quantisation followed by dequantisation, which destroys the multiplicative training history), or that the measurement itself is flawed (for example by including bits below the noise floor in the regression). We use the law to triage measurements before incorporating them into the FCN evidence register.

12.2 The law as a closed-form anchor for the ceiling formula

The ceiling formula derived in the adjacent Benford-theory entry depends on the mantissa entropy, which in turn depends on the bit-level biases captured by the $r = 2$ law. The closed-form sequence $|b_i| = |B_0| \cdot 2^{i-M+1}$ gives the mantissa entropy directly, without requiring a separate empirical estimate, and the empirical regression on a single trained model serves to verify that the closed form applies. The law therefore acts as a closed-form anchor that fixes one of the three terms in the ceiling formula and reduces the architecture-dependent variability to a single quantity.

12.3 The law and the cross-model delta regime

The cross-model delta regime breaks the $r = 2$ law on the mantissa of the residual, as documented in the adjacent Fine-Tuning Conservatism Law entry. The delta mantissa is not the asymptotic product of training updates; it is the small difference between two such products. The Benford derivation does not apply, and the bit-level bias sequence of the residual is concentrated at low bit positions rather than spread geometrically from the most significant mantissa bit. This Benford-breaking is the mechanism by which the cross-model delta ceiling exceeds the single-model ceiling.

12.4 The law and the FP32 ceiling

The FP32 ceiling reported in the adjacent FP32 Ceiling entry is the format-width analogue of the BF16 ceiling. Under the $r = 2$ law, the FP32 mantissa has approximately twelve bits above the noise floor and approximately eleven bits below. The eleven below-noise-floor bits contribute one bit of incompressible noise each, raising the FP32 mantissa entropy to approximately twenty-two out of twenty-three bits. The $r = 2$ law thus directly predicts the FP32 ceiling up to the architecture-dependent exponent term. The empirical FP32 ceiling of approximately $1.18\times$ is consistent with this prediction.

12.5 The law as a diagnostic of training maturity

The position of the noise floor within the mantissa is a function of the number of mantissa bits above the detection threshold for the $r = 2$ law. A network trained for fewer steps has a smaller cumulative multiplicative compounding, a smaller deviation from the initialisation distribution, and (in some training regimes) a noise floor that sits higher in the mantissa. A network trained for many more steps has a smaller noise floor and more bits above it. The $r = 2$ law therefore provides an indirect diagnostic of training maturity: more bits above the noise floor implies more training. We do not in this paper develop a formal calibration of this diagnostic; we note it as a consequence of the law.

12.6 The law and the universal constant

The constant $|B_0| = \log_2 3 - 3/2 \approx 0.084962500721156$ appears in the most-significant mantissa bit bias as a consequence of the binary Benford density integrated over the half-interval $[1, 3/2)$. The value $\log_2 3$ has independent significance: it is the entropy of a fair three-symbol source and the Hausdorff dimension of the Sierpinski triangle. The appearance of $\log_2 3$ in the trained-weight context is not, however, a coincidence with these other appearances; it is a consequence of the IEEE 754 mantissa interval being $[1, 2)$ with the natural midpoint at $3/2$. The constant $|B_0|$ is thus the binary-mantissa analogue of similar Benford constants that appear under other base choices, but the specific value is set by the binary structure of the floating-point format.

12.7 Sign and significance interpretation

The sign of B_0 as defined in this paper, $B_0 = \log_2 3 - 3/2 > 0$, follows the convention that B_0 is the magnitude of the bias and the most-significant mantissa bit is biased toward zero. This convention reflects that the binary Benford density is concentrated near the lower edge of the significand interval $[1, 2)$, so the most-significant mantissa bit is more often zero than one. The bias is therefore negative in the signed sense, $b_{M-1} = -B_0 < 0$, with $P(\text{bit } M - 1 = 1) = 2 - \log_2 3 \approx 0.41504$.

Earlier internal notes in the FCN evidence register used the positive sign $b_{M-1} = +B_0$. This was an error that we record explicitly because it propagated into early derivations of the ceiling formula. The corrected convention is the one used throughout this paper and is what aligns with the empirical measurement: the most-significant mantissa bit is biased toward zero, not toward one.

13 Discussion

The result is interpreted as a property of the trained-weight distribution and not as a property of any single codec or compression pipeline. It is also a property of a regime: the asymptotic, noise-free portion of the mantissa for a mature, fully-trained checkpoint. Outside that regime, the law neither holds nor fails; it does not apply.

We emphasise that the $r = 2$ law is a consequence of Benford, not an independent regularity. The Benford derivation gives an exact closed-form sequence of biases whose ratio converges to two from below as the bit index decreases. The law is therefore best understood as the leading-order behaviour of that closed-form sequence. Where the closed form differs from the law, the closed form is correct; the law captures only the asymptotic ratio.

The law gives a quantitative way to predict the mantissa entropy of a trained model from a single constant. For BF16, this is $H(\text{mantissa}) \approx 6.97$ bits out of seven; for FP32, the corresponding figure is $H(\text{mantissa}) \approx 22.33$ bits out of twenty-three. These values are recorded in adjacent FCN entries

and are corroborated by the bit-level biases reported here.

13.1 Connection to the noise floor

The law is a quantitative statement about the bias of each mantissa bit, but it also implicitly defines the noise floor of a trained network. The detection threshold for $|b_i|$ on a sample of size N values is approximately $1/\sqrt{N}$ in absolute magnitude (the standard error of a binomial bit-frequency estimator on equally likely outcomes). The law predicts $|b_i| = |B_0| \cdot 2^{i-M+1}$, so a bit position i is detectable above the noise floor when

$$|B_0| \cdot 2^{i-M+1} \gtrsim \frac{1}{\sqrt{N}}, \quad (13)$$

or equivalently

$$i \gtrsim M - 1 - \log_2(|B_0|\sqrt{N}). \quad (14)$$

For $N \approx 10^9$ values (the size of a typical large language model parameter count), this gives $i \gtrsim M - 1 - 11 \approx M - 12$. For BF16 ($M = 7$), this places the noise floor near $i = -5$, which is below the lowest bit ($i = 0$); all seven mantissa bits should therefore carry detectable bias on a model of that size. For FP32 ($M = 23$), the predicted noise floor is near $i = 11$, consistent with the empirical observation that approximately twelve mantissa bits carry detectable bias on GPT-2 XL.

13.2 Predicting the per-bit entropy contribution

The mantissa entropy contribution of bit position i is, to leading order in $|b_i|$,

$$H_i \approx 1 - \frac{1}{2 \ln 2} \cdot (2|b_i|)^2 = 1 - \frac{2|b_i|^2}{\ln 2}. \quad (15)$$

Substituting $|b_i| = |B_0| \cdot 2^{i-M+1}$ gives

$$H_i \approx 1 - \frac{2|B_0|^2}{\ln 2} \cdot 4^{i-M+1}. \quad (16)$$

The deficit relative to one bit decreases geometrically with bit position, halving at every step downwards from the most significant mantissa bit. Summing across all mantissa bits gives a total mantissa entropy of

$$H(\text{mantissa}) \approx M - \frac{2|B_0|^2}{\ln 2} \cdot \sum_{i=0}^{M-1} 4^{i-M+1} \approx M - \frac{2|B_0|^2}{\ln 2} \cdot \frac{4}{3} \cdot 1 = M - \frac{8|B_0|^2}{3 \ln 2}. \quad (17)$$

For $M = 7$ and $|B_0| = 0.08496$, this gives $H(\text{mantissa}) \approx 7 - 0.028 = 6.972$ bits, recovering the Benford value. The $r = 2$ law thus directly predicts the mantissa entropy of a trained network to within the precision required by the ceiling formula.

14 Limitations

We note five limitations.

1. The law is asymptotic. Below the training noise floor, the measured bias is dominated by stochastic optimisation noise and is consistent with a fair coin. The fit range must exclude these positions, and we do not claim the law holds below the noise floor.

2. The law is derived under the assumption that the significand distribution approaches Benford over $[1, 2)$. Distributions that violate Benford on that interval, for example a uniform weight distribution on $[-1, 1]$, do not exhibit the law. We do not claim universality across all numerical data.
3. The empirical evidence is drawn from a finite set of public-model artefacts. We do not claim the law holds for every trained network. We claim that the law holds for the models tested and that the derivation gives a mechanistic reason to expect it to hold for SGD-trained networks more generally.
4. The closed-form Benford sequence shows residual deviations from a pure geometric law near the most-significant mantissa bit (Table 1). These deviations are real and quantitative; the geometric law is correct only in the leading-order asymptotic sense.
5. Public reproducibility is constrained by disclosure control. The internal verification artefacts that produced the measured ratios are maintained by Forlais Group but are not included in this paper.

15 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here. The public-model artefacts used in the measurements are released by their respective developers under their stated licences.

16 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

17 Summary of the analytical and empirical claims

In order, the analytical and empirical claims of this paper are: (i) the closed-form Benford bias sequence on the IEEE 754 mantissa is exact; (ii) the asymptotic geometric ratio of consecutive biases is exactly two; (iii) the absolute scale of the sequence is set by $|B_0| = \log_2 3 - 3/2$ at the most-significant mantissa bit; (iv) BF16 Mistral-7B yields $r = 2.014$ with $R^2 = 0.9997$ over bit positions m_3 through m_6 ; (v) FP32 GPT-2 XL yields $r = 1.995$ with $R^2 = 0.9998$ over bit positions m_{11} through m_{22} ; (vi) the law is asymptotic and does not apply below the training noise floor; (vii) the law is format-independent; (viii) the law is scale-independent; (ix) the cross-architecture data on five public-model artefacts are consistent with the law within a few per cent; and (x) the law functions as an analytical anchor for the ceiling formula derived in the adjacent Benford-theory entry.

18 Conclusion

We report the $r = 2$ law for mantissa-bit bias in trained neural network weights. For a fixed IEEE 754 binary format with M mantissa bits, the bias of mantissa bit i above the training noise floor obeys $b_i \approx B_0 \cdot 2^{i-M+1}$, with $B_0 = \log_2 3 - 3/2$. We verify the law on BF16 weights of Mistral-7B at $r = 2.014$ with $R^2 = 0.9997$ and on FP32 weights of GPT-2 XL at $r = 1.995$ with $R^2 = 0.9998$. The law follows from Benford's law applied to the binary mantissa interval and is best understood as the leading-order asymptotic of an exact closed-form bias sequence whose consecutive ratios converge to two from below. The result is reported in its public-safe form and should be read alongside the limitations enumerated above.

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Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
1.0	14 May 2026	Full evidence-grounded manuscript with analytical derivation, exact Benford sequence, empirical regression results, cross-architecture comparison, and connection to public literature.	Oliver Cooper